

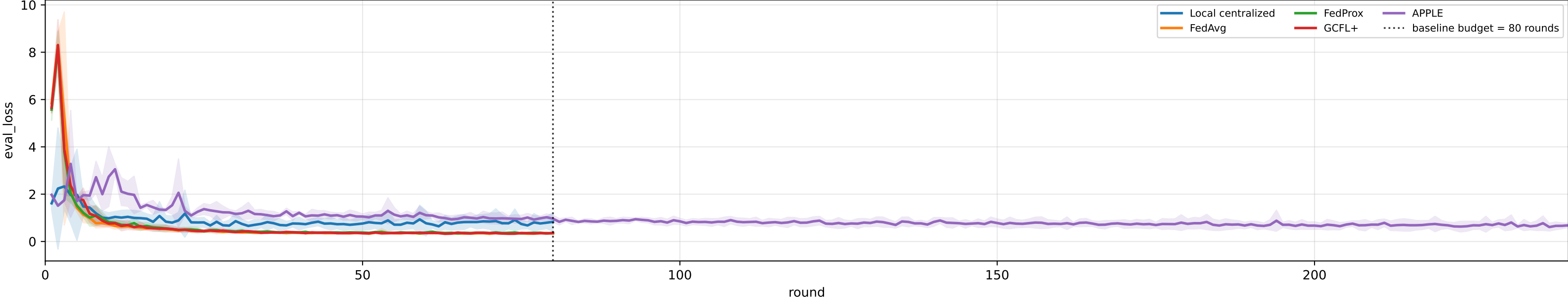
Controlled q-label heterogeneity: iid | aggregated / pooled performance over 5 clients |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

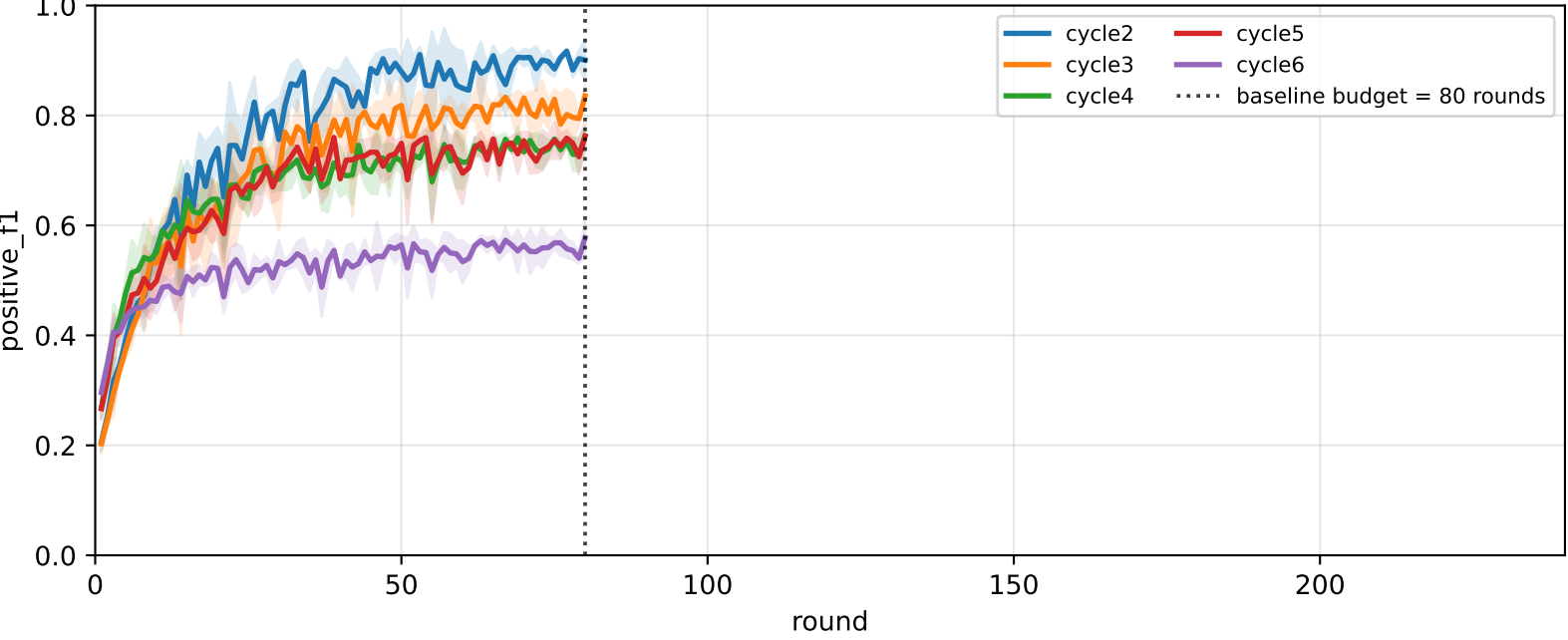
Best logged test performance (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	64.2% ± 0.6%	67.8% ± 0.9%	79.6% ± 4.0%	78.3% ± 5.1%	66.8% ± 1.9%	61.9% ± 2.0%	48.9% ± 0.2%
FedAvg	70.0% ± 2.3%	72.5% ± 2.2%	86.3% ± 3.0%	80.4% ± 4.9%	71.2% ± 1.5%	67.1% ± 4.1%	57.6% ± 3.6%
FedProx	65.8% ± 1.0%	68.8% ± 0.6%	83.6% ± 2.3%	77.3% ± 1.3%	67.3% ± 3.8%	63.0% ± 0.4%	52.7% ± 1.5%
GCFL+	67.0% ± 0.9%	70.1% ± 0.8%	81.8% ± 1.2%	82.6% ± 1.2%	69.0% ± 2.3%	64.1% ± 2.7%	52.8% ± 0.7%
APPLE	68.2% ± 4.9%	70.4% ± 4.6%	82.2% ± 11.5%	74.8% ± 11.0%	71.1% ± 1.1%	65.5% ± 6.4%	58.0% ± 4.0%

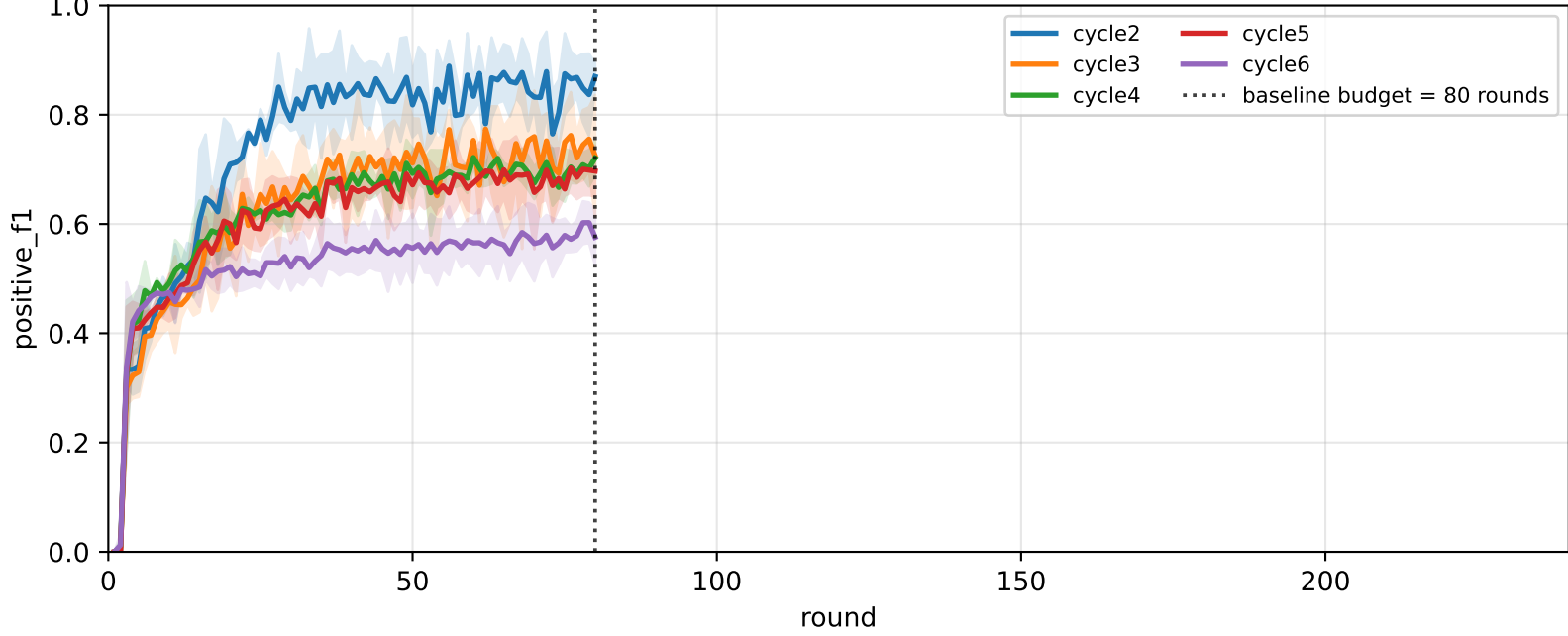
Validation loss comparison (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)



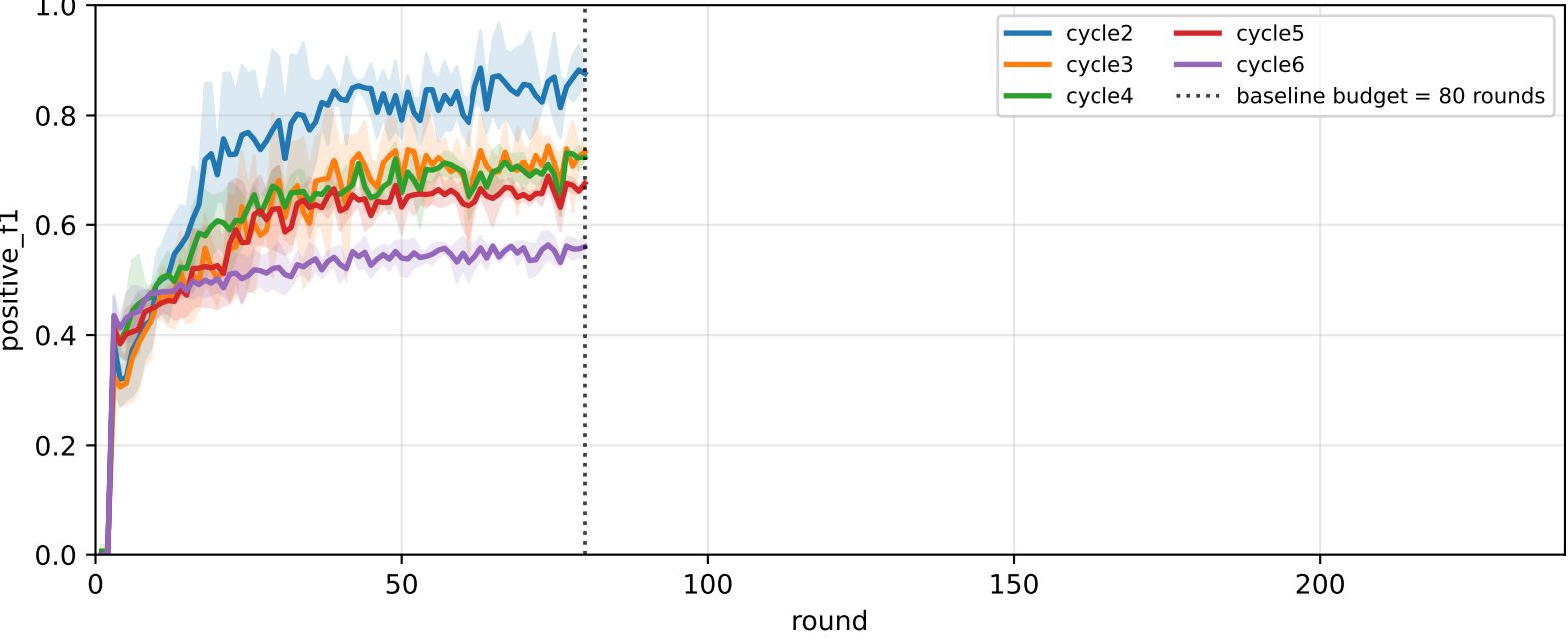
Pooled local centralized positive F1



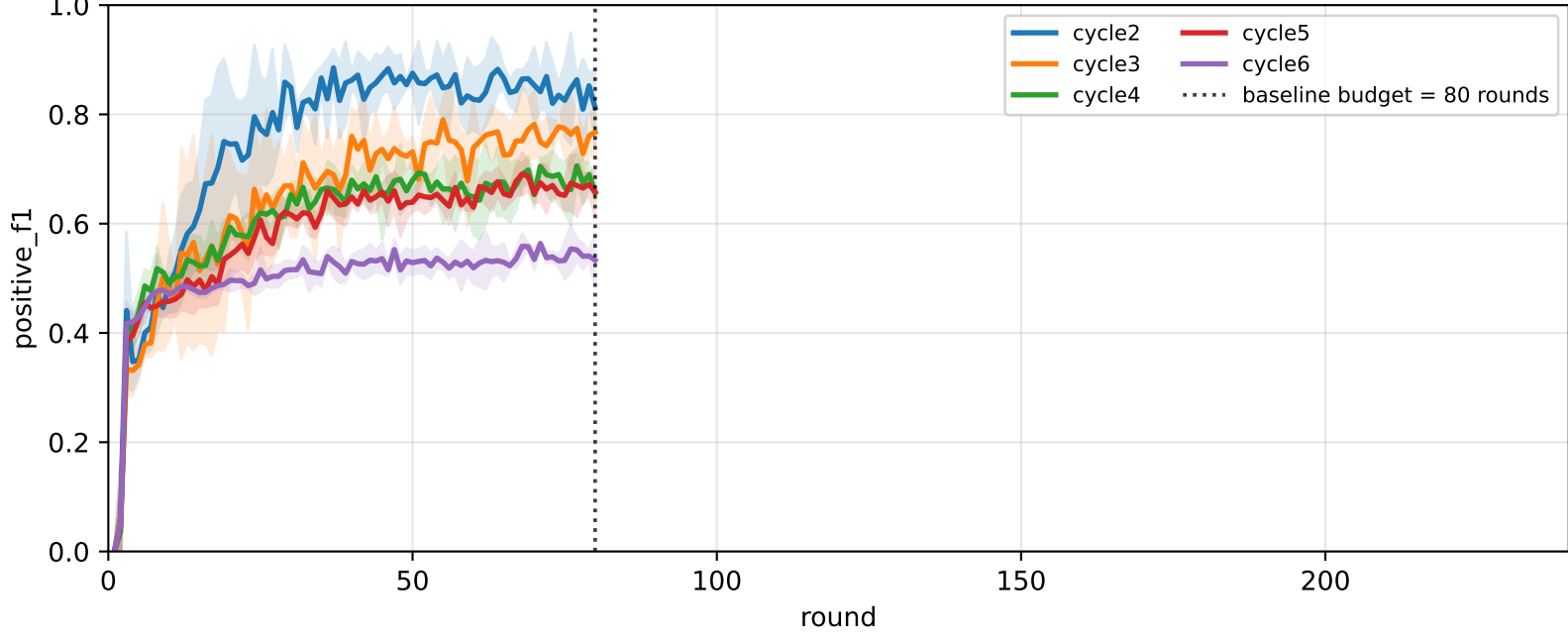
Pooled FedAvg positive F1



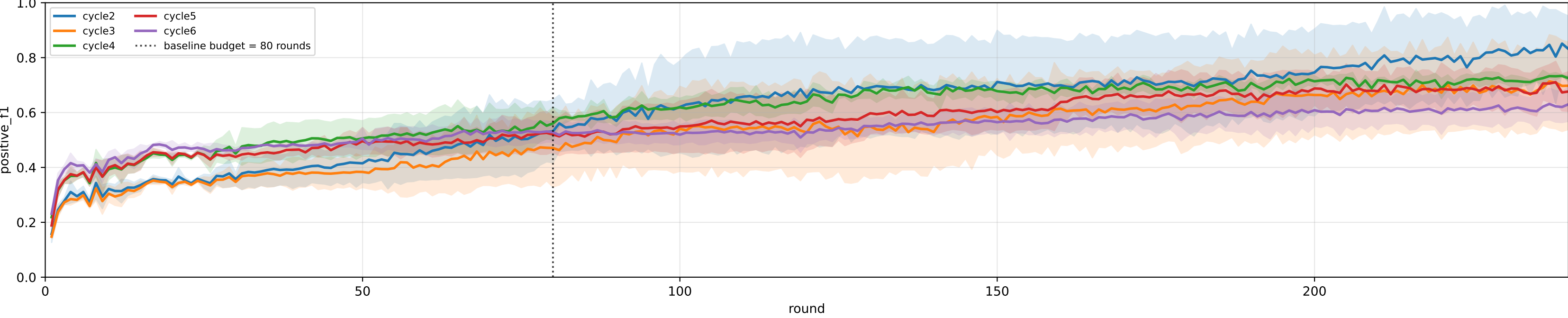
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, equal 240-round budget)



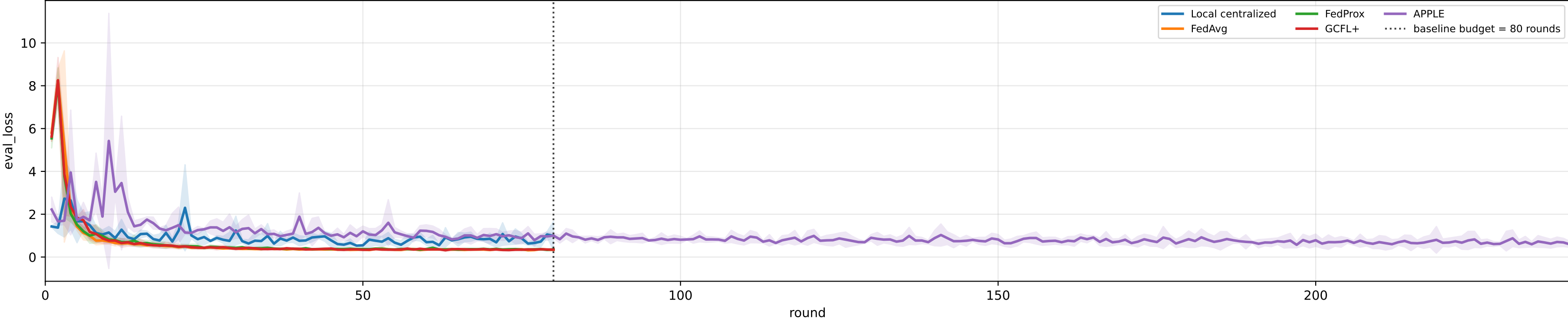
Controlled q-label heterogeneity: iid | family aggregated / pooled performance |
family=q_task_cycle2 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

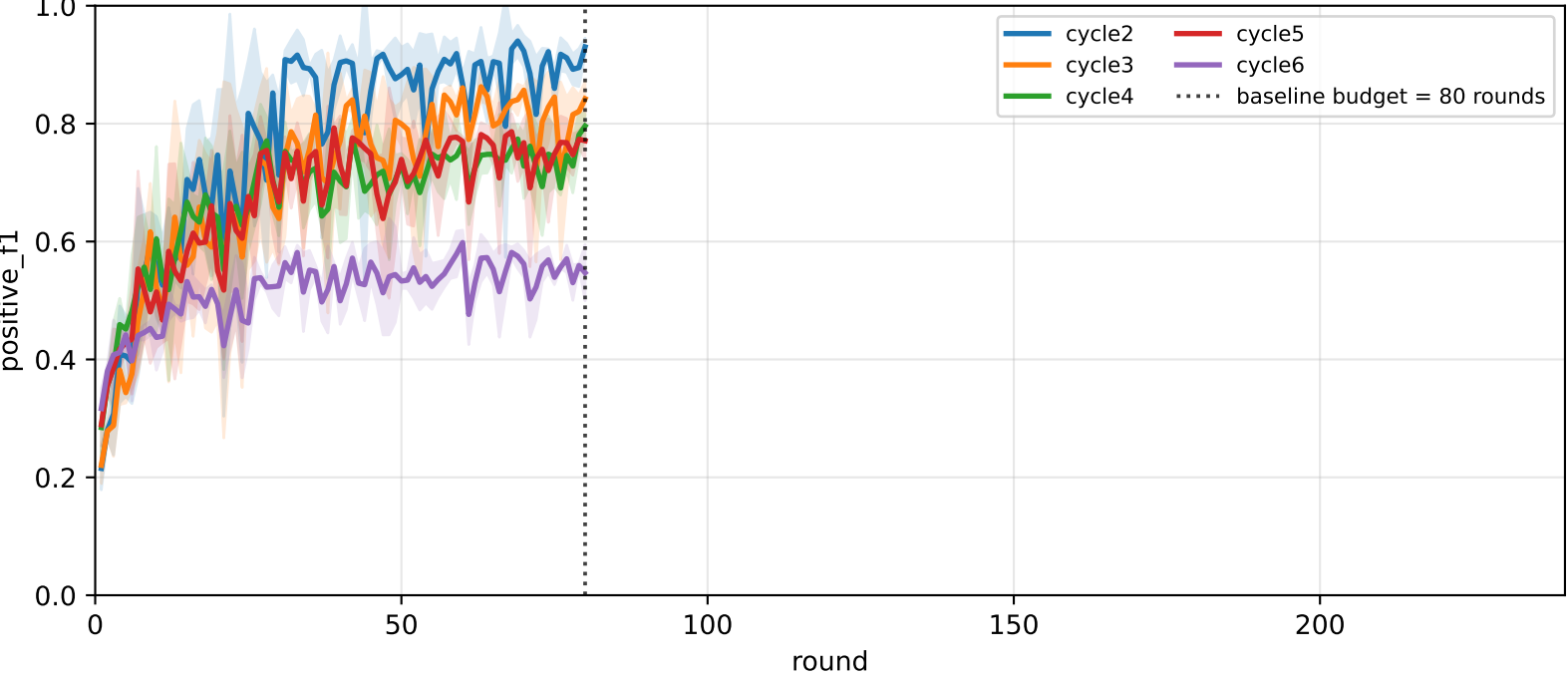
Best logged test performance (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	61.7% ± 2.9%	65.8% ± 2.5%	83.4% ± 5.2%	77.4% ± 2.3%	64.8% ± 2.9%	56.1% ± 5.8%	47.3% ± 2.8%
FedAvg	70.1% ± 2.2%	72.8% ± 2.1%	87.4% ± 4.0%	81.0% ± 5.5%	70.6% ± 1.5%	67.3% ± 4.2%	57.8% ± 3.7%
FedProx	66.1% ± 1.0%	69.0% ± 0.5%	84.1% ± 2.9%	77.5% ± 1.7%	67.1% ± 3.2%	63.5% ± 0.1%	52.9% ± 1.7%
GCFL+	67.7% ± 0.7%	70.6% ± 0.6%	81.9% ± 2.1%	82.9% ± 1.8%	69.8% ± 1.9%	65.0% ± 2.2%	53.4% ± 0.9%
APPLE	68.2% ± 4.2%	70.8% ± 3.8%	84.1% ± 12.3%	75.0% ± 11.3%	72.5% ± 0.8%	65.1% ± 6.6%	57.2% ± 3.0%

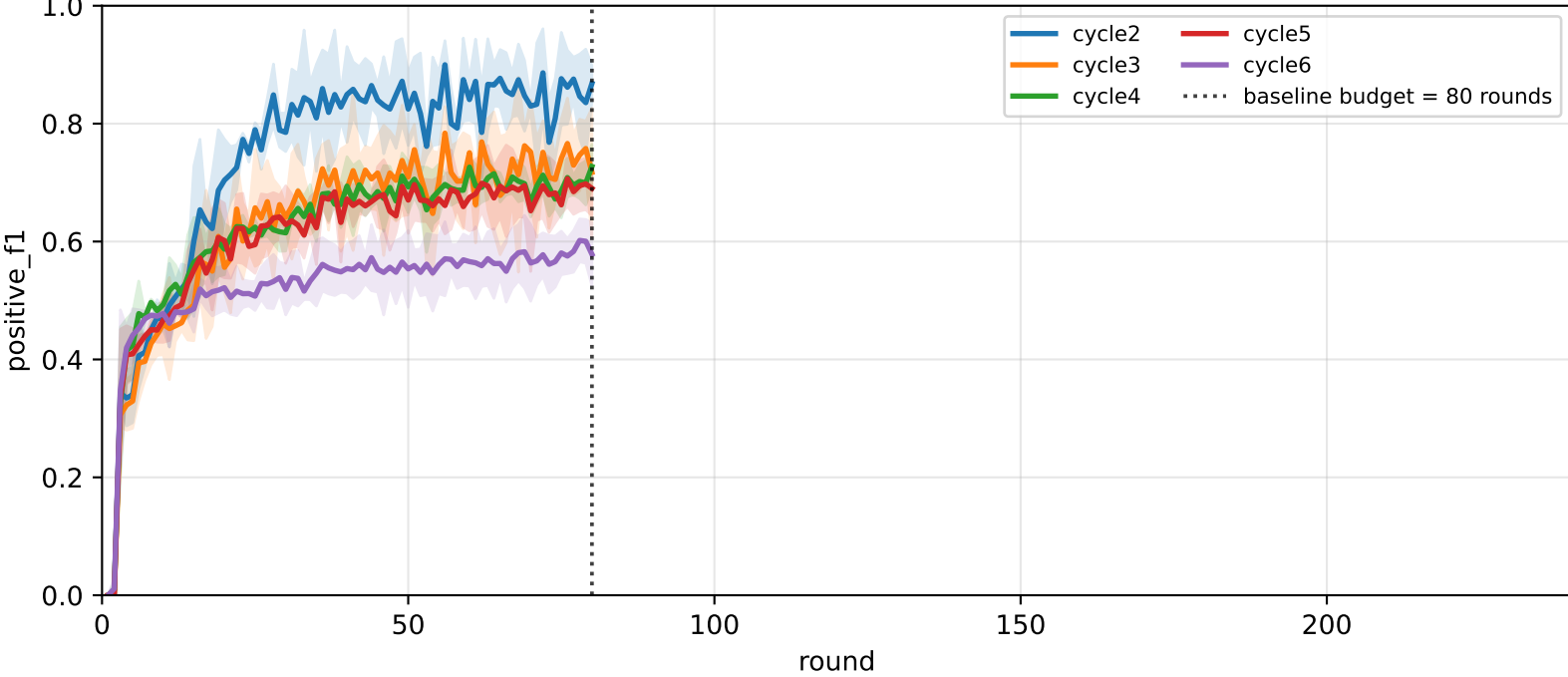
Validation loss comparison (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)



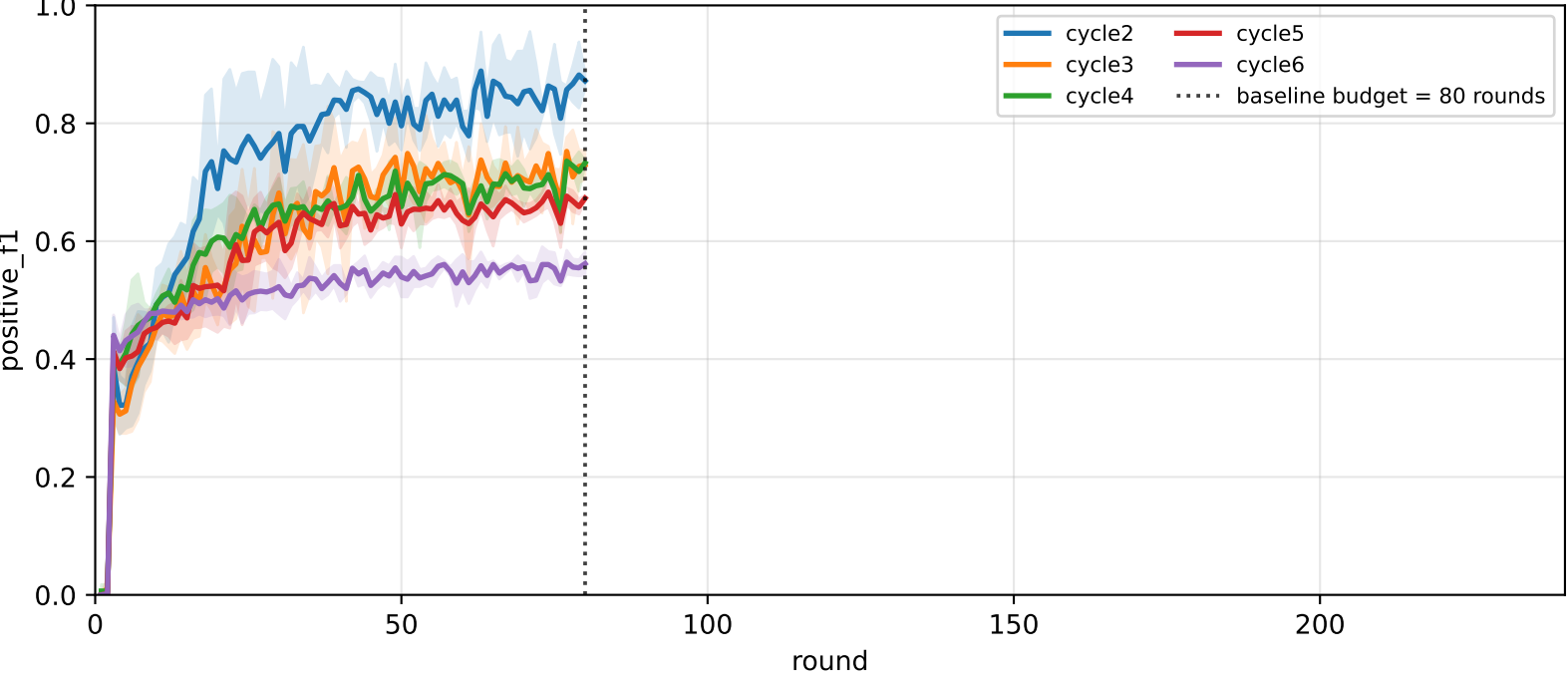
Pooled local centralized positive F1



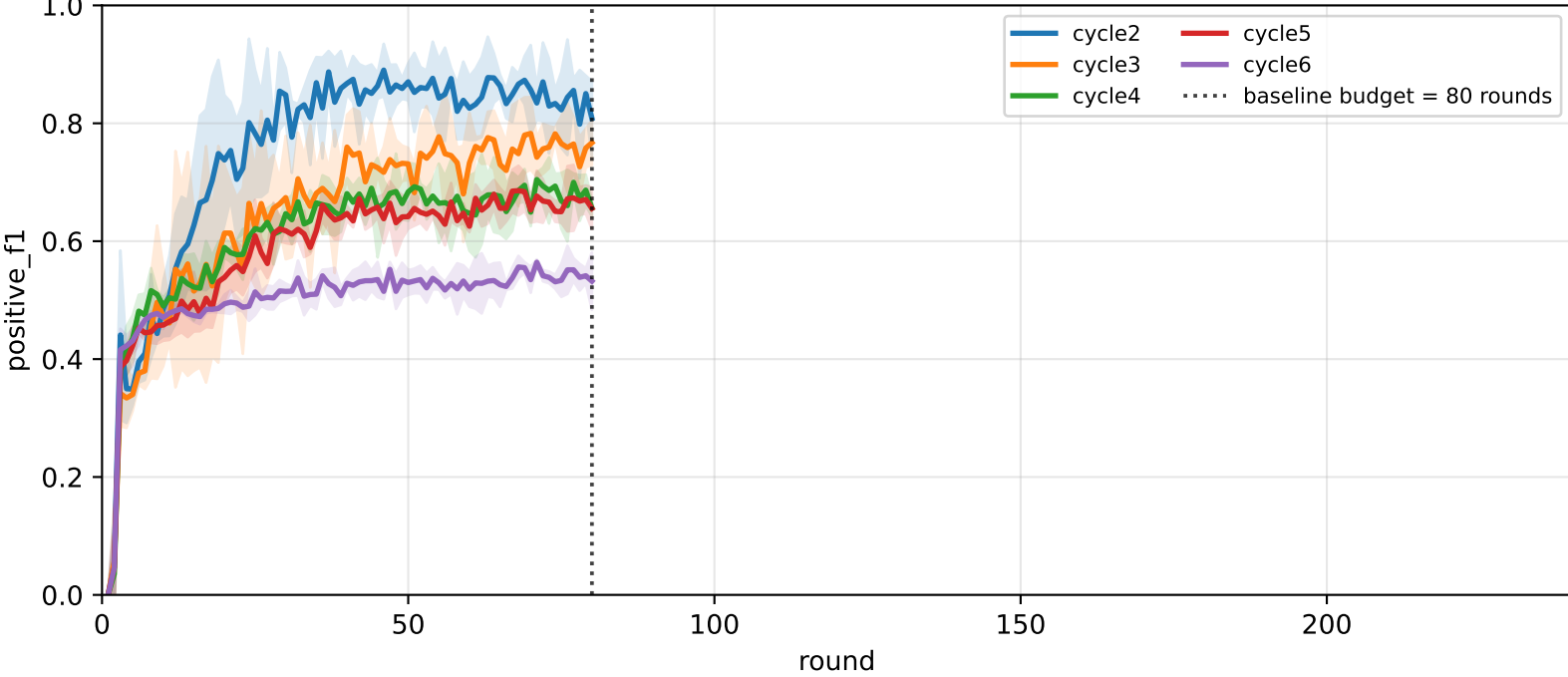
Pooled FedAvg positive F1



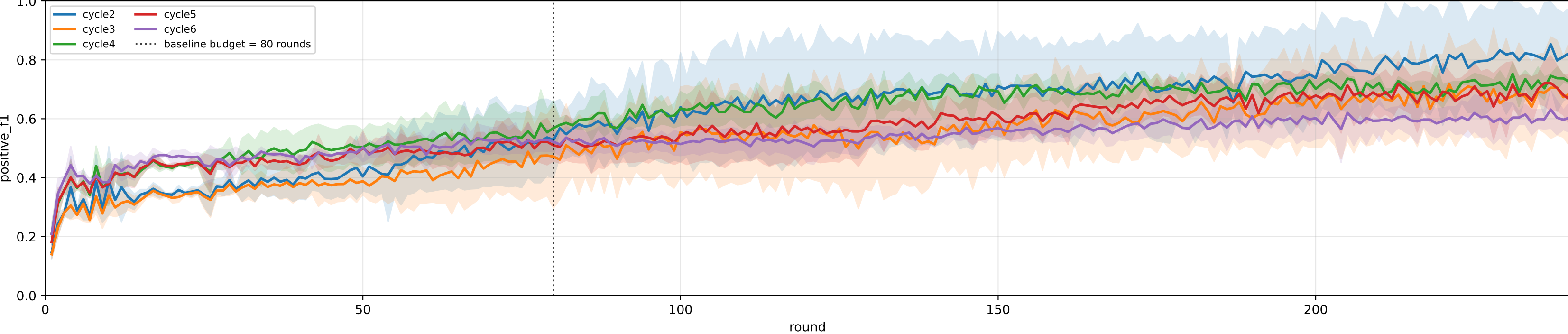
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, equal 240-round budget)



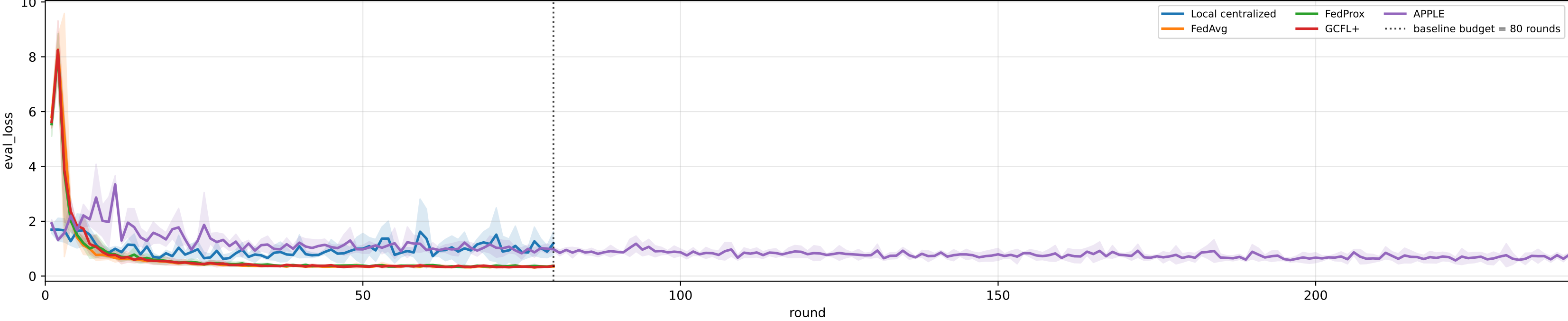
Controlled q-label heterogeneity: iid | family aggregated / pooled performance |
family=q_task_cycle3 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

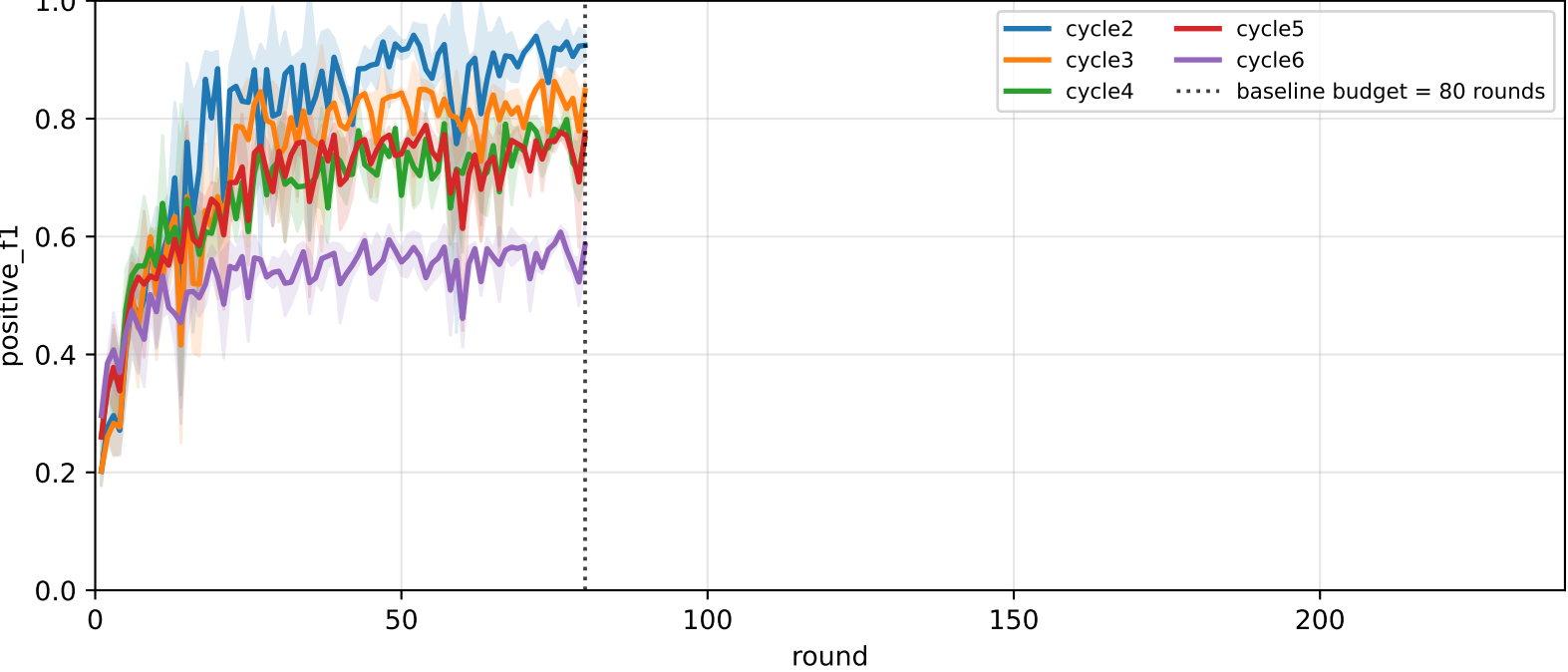
Best logged test performance (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	59.3% ± 1.8%	62.3% ± 1.2%	70.6% ± 12.2%	70.7% ± 7.8%	63.8% ± 2.1%	59.4% ± 6.7%	46.7% ± 4.1%
FedAvg	70.0% ± 2.1%	72.6% ± 1.7%	86.7% ± 2.6%	80.6% ± 4.2%	71.3% ± 2.5%	66.8% ± 3.9%	57.7% ± 4.0%
FedProx	66.0% ± 0.6%	69.1% ± 0.3%	84.0% ± 2.3%	78.2% ± 1.5%	67.6% ± 3.3%	62.9% ± 0.9%	52.6% ± 1.5%
GCFL+	67.0% ± 1.0%	70.1% ± 1.0%	81.4% ± 1.8%	83.2% ± 0.1%	69.4% ± 1.9%	63.7% ± 2.8%	52.6% ± 0.7%
APPLE	69.8% ± 5.2%	71.7% ± 4.5%	82.3% ± 9.6%	75.1% ± 11.3%	72.5% ± 1.1%	68.6% ± 7.2%	60.0% ± 5.0%

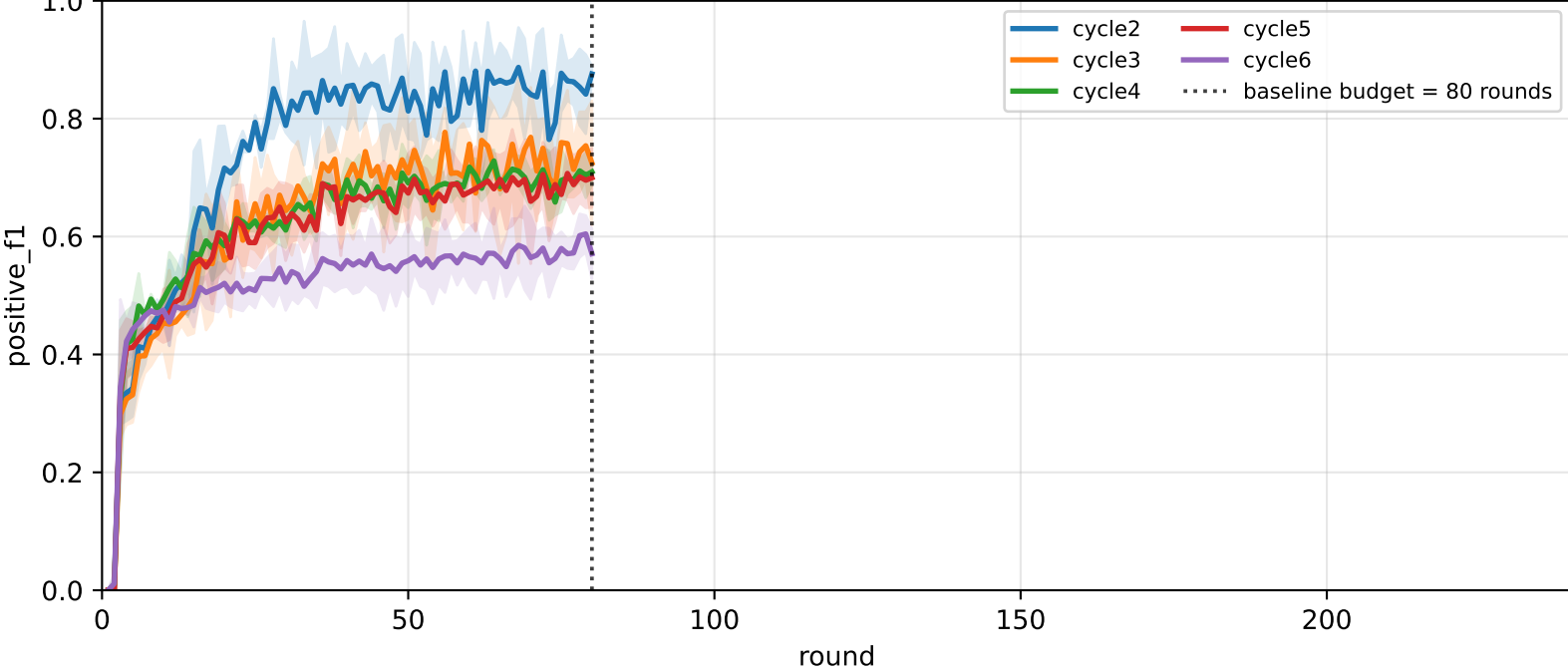
Validation loss comparison (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)



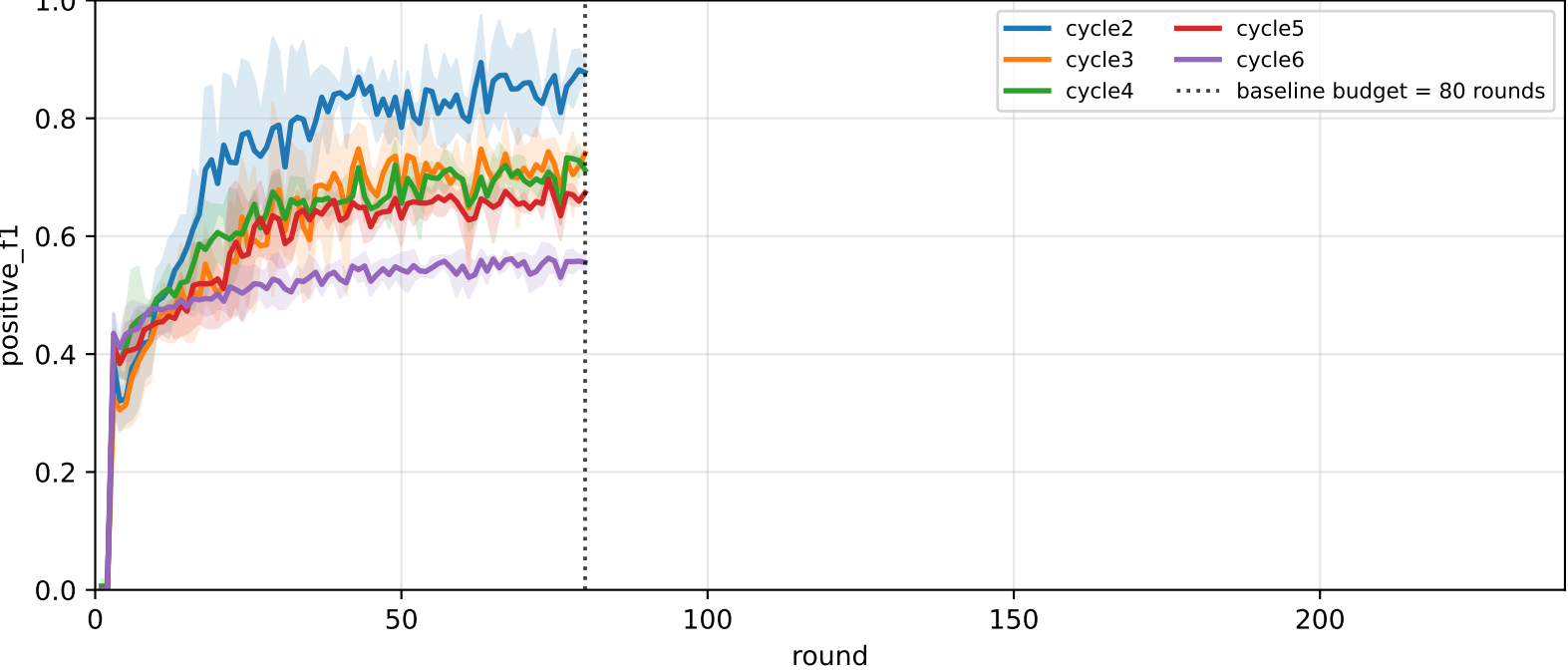
Pooled local centralized positive F1



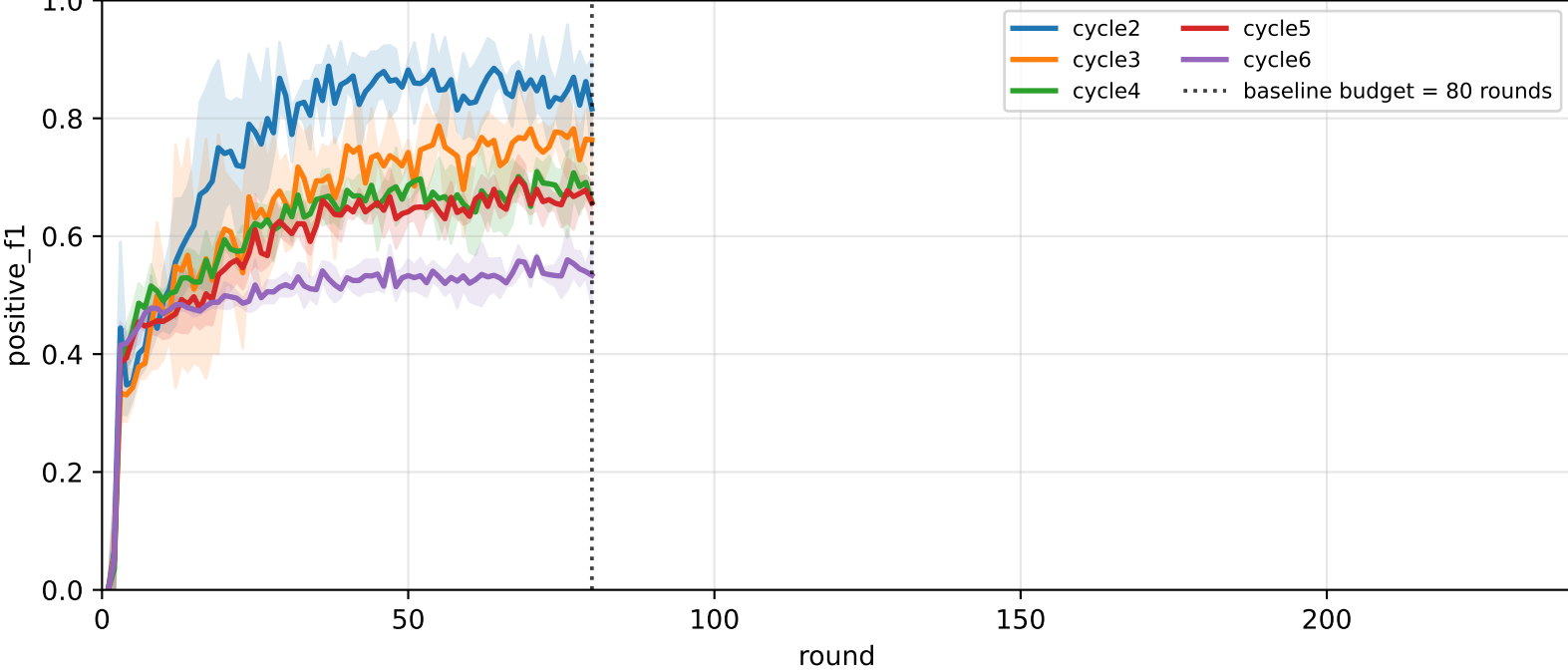
Pooled FedAvg positive F1



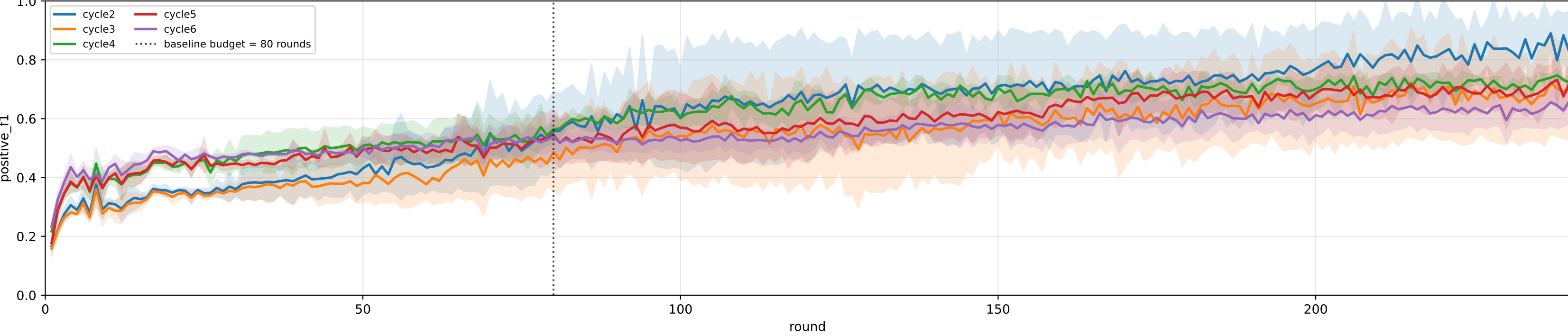
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, equal 240-round budget)



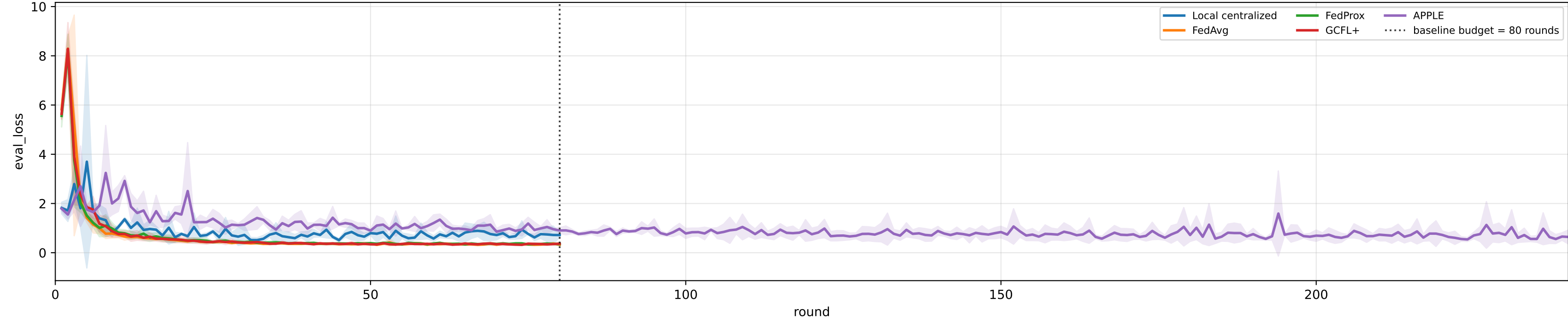
Controlled q-label heterogeneity: iid | family aggregated / pooled performance |
family=q_task_cycle4 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

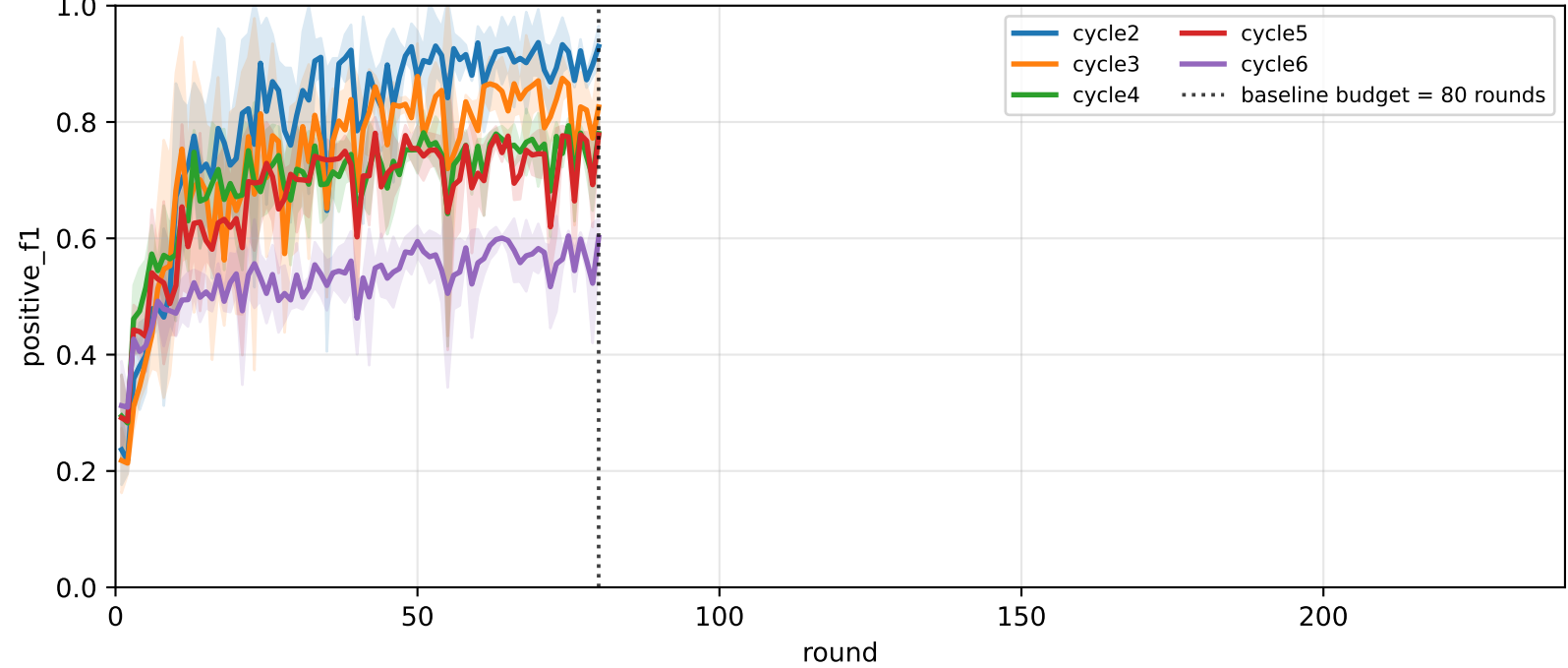
Best logged test performance (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	65.1% ± 6.6%	68.5% ± 6.7%	78.1% ± 11.0%	82.9% ± 5.3%	69.7% ± 5.5%	62.2% ± 8.5%	49.8% ± 5.3%
FedAvg	70.1% ± 2.1%	72.8% ± 1.9%	86.8% ± 2.8%	80.9% ± 4.8%	71.6% ± 1.4%	67.1% ± 4.4%	57.4% ± 3.2%
FedProx	65.7% ± 0.7%	68.6% ± 0.4%	83.2% ± 2.8%	77.5% ± 1.3%	66.9% ± 3.3%	62.7% ± 0.6%	52.8% ± 1.3%
GCFL+	66.5% ± 1.2%	69.5% ± 0.9%	80.9% ± 0.7%	81.7% ± 1.6%	68.7% ± 2.9%	63.2% ± 3.2%	52.8% ± 1.0%
APPLE	67.2% ± 5.2%	69.2% ± 4.9%	80.7% ± 11.3%	73.4% ± 10.5%	69.3% ± 1.6%	64.6% ± 6.7%	57.9% ± 4.6%

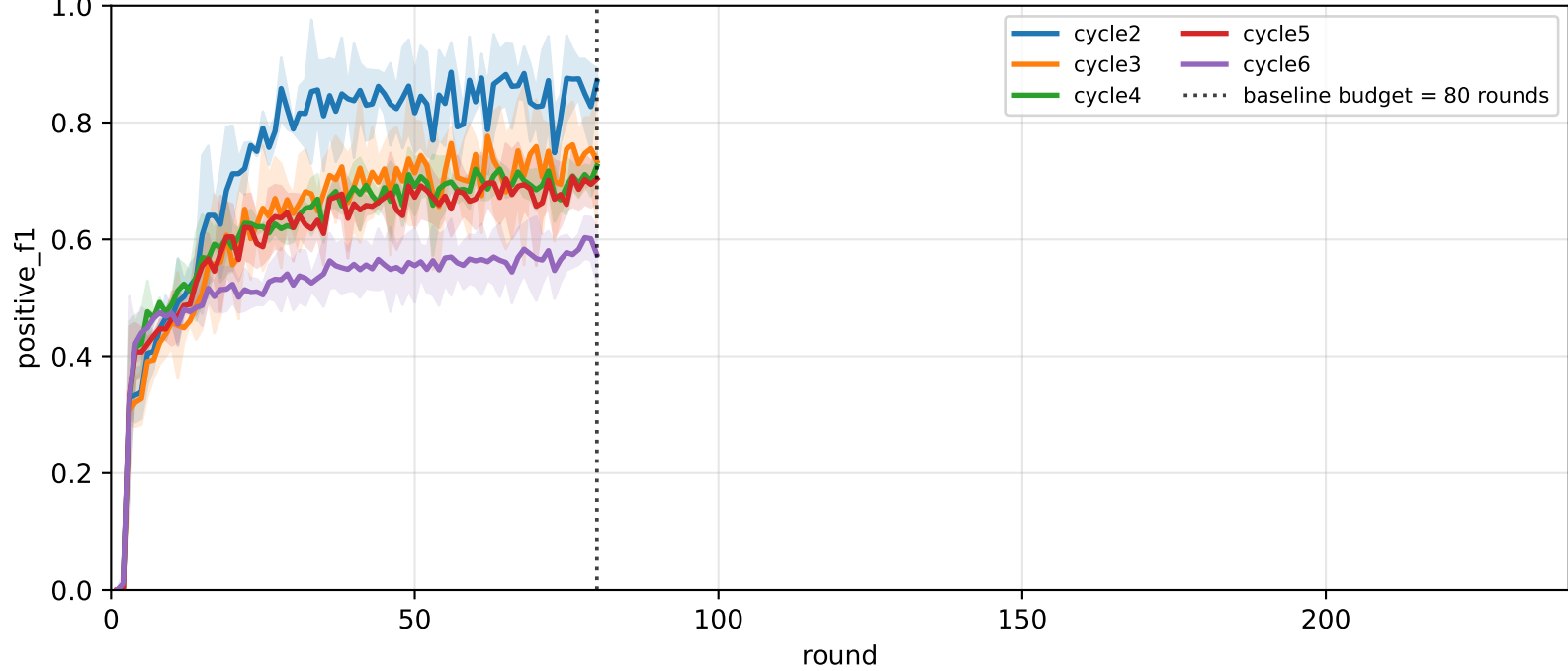
Validation loss comparison (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)



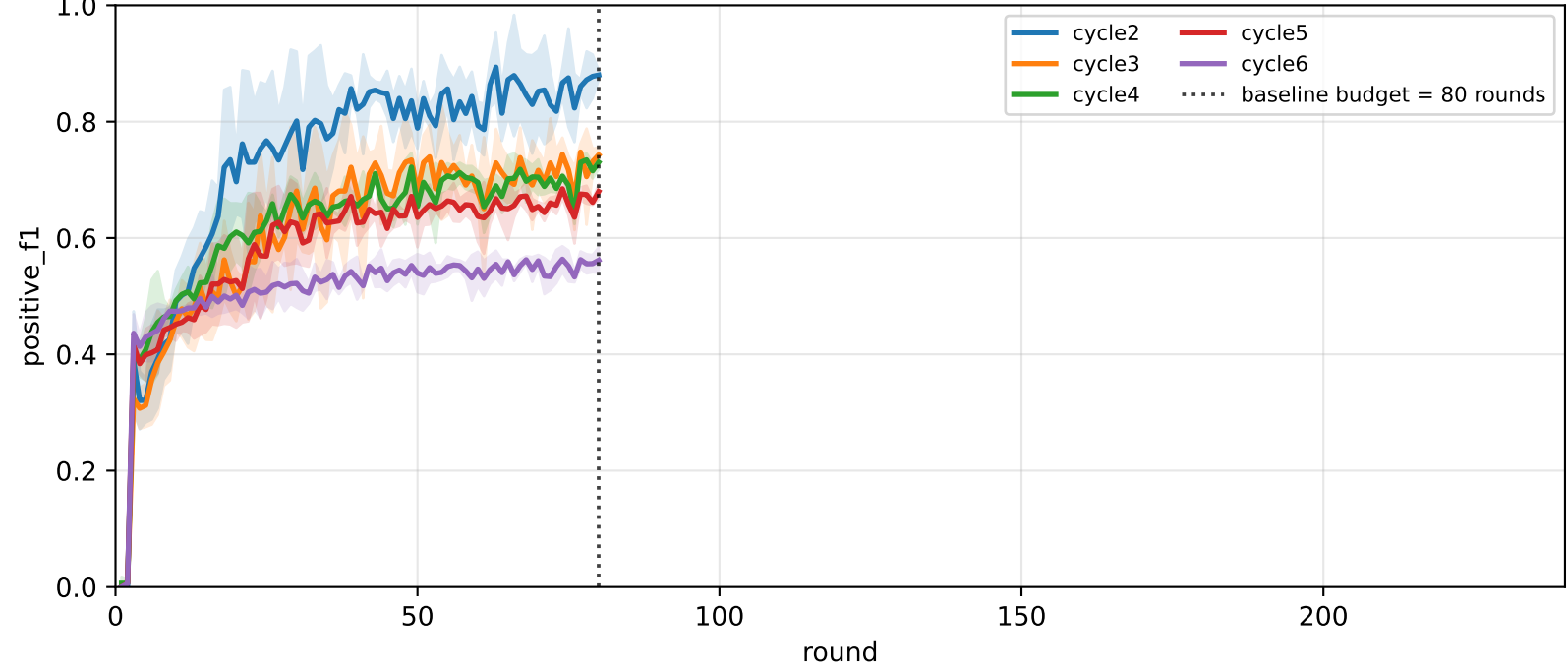
Pooled local centralized positive F1



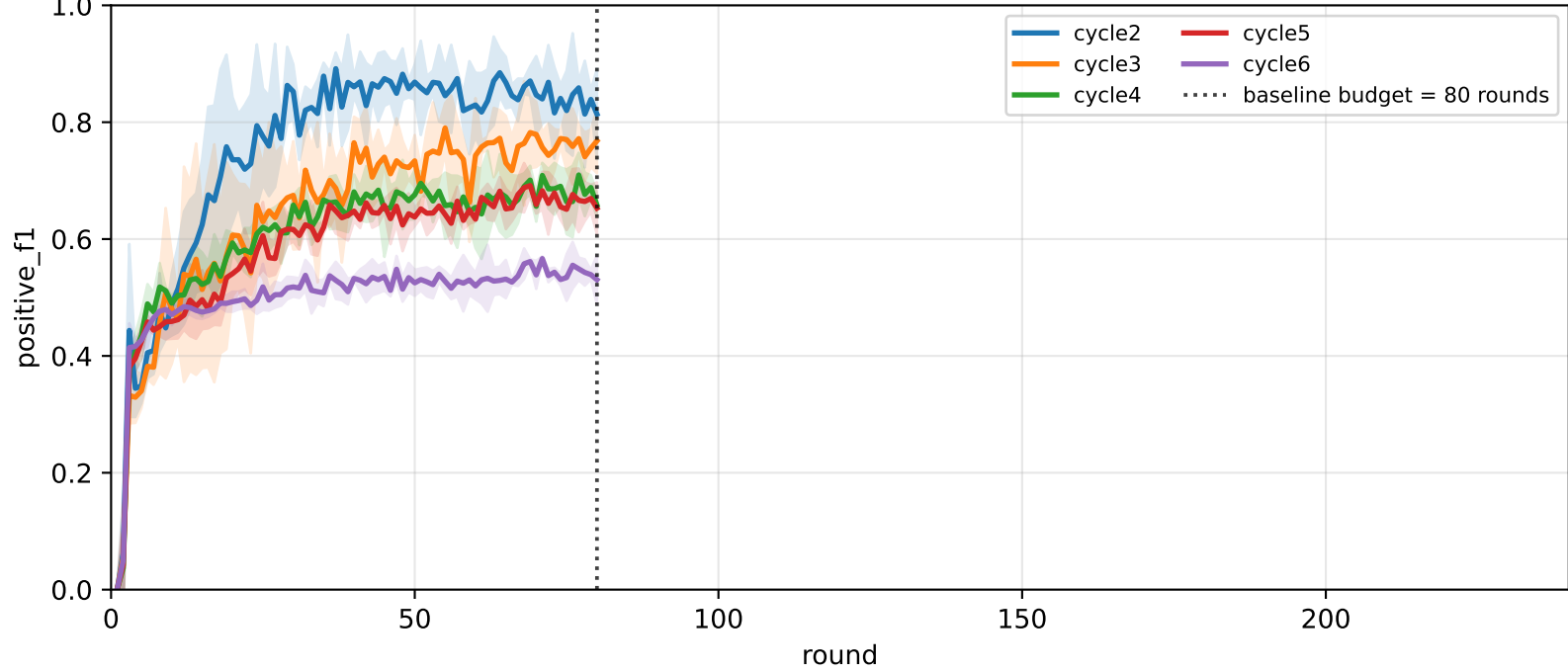
Pooled FedAvg positive F1



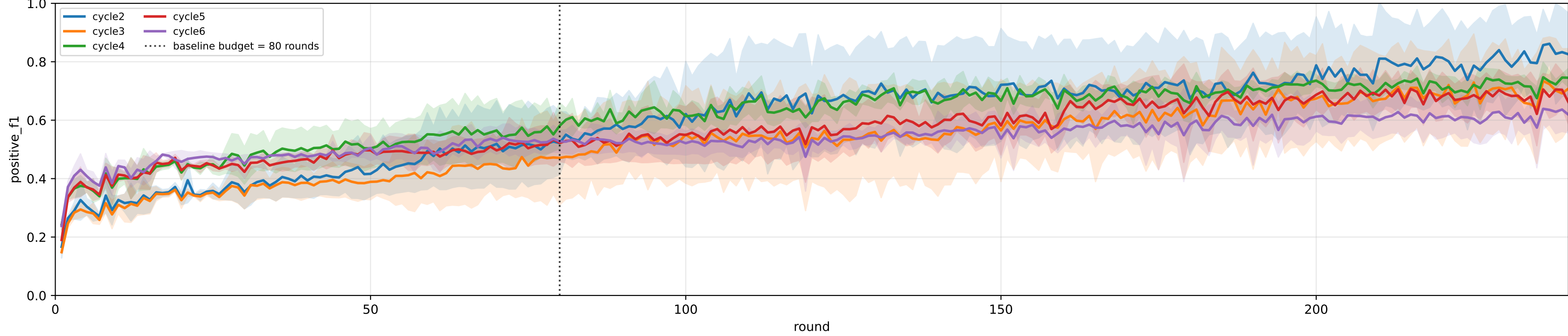
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, equal 240-round budget)



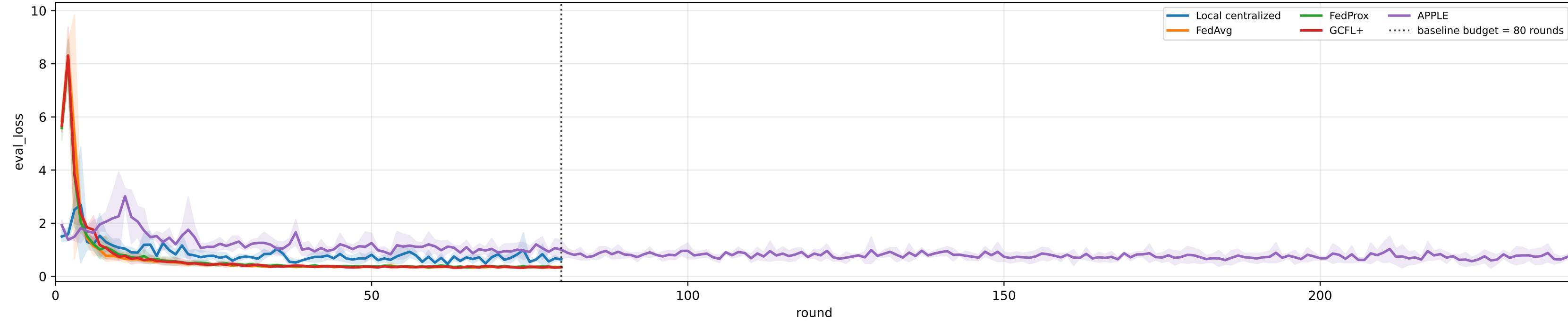
Controlled q-label heterogeneity: iid | family aggregated / pooled performance |
family=q_task_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

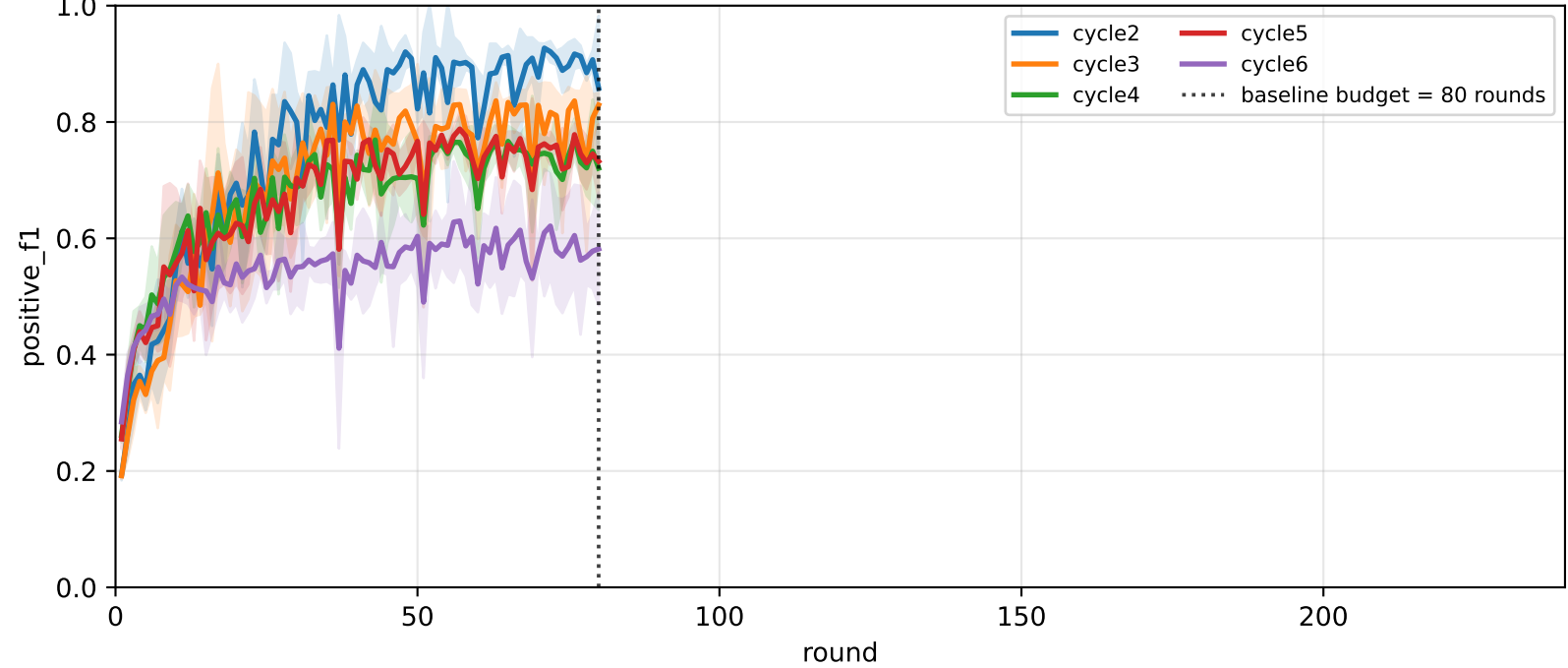
Best logged test performance (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	66.7% ± 3.2%	69.8% ± 3.5%	83.1% ± 3.5%	76.9% ± 8.5%	67.2% ± 1.6%	69.7% ± 4.1%	52.1% ± 4.2%
FedAvg	69.5% ± 2.6%	72.0% ± 2.5%	84.7% ± 2.6%	79.6% ± 5.0%	71.2% ± 1.4%	66.8% ± 4.1%	57.5% ± 3.7%
FedProx	65.4% ± 1.1%	68.3% ± 0.7%	82.8% ± 2.3%	76.6% ± 1.2%	67.0% ± 4.5%	62.5% ± 0.3%	52.5% ± 1.4%
GCFL+	67.1% ± 0.8%	70.2% ± 0.5%	82.3% ± 1.4%	82.4% ± 1.7%	68.8% ± 1.6%	64.7% ± 3.1%	52.6% ± 0.7%
APPLE	67.8% ± 5.5%	70.4% ± 5.6%	84.3% ± 13.0%	74.3% ± 10.4%	72.9% ± 4.7%	64.2% ± 6.3%	56.4% ± 4.8%

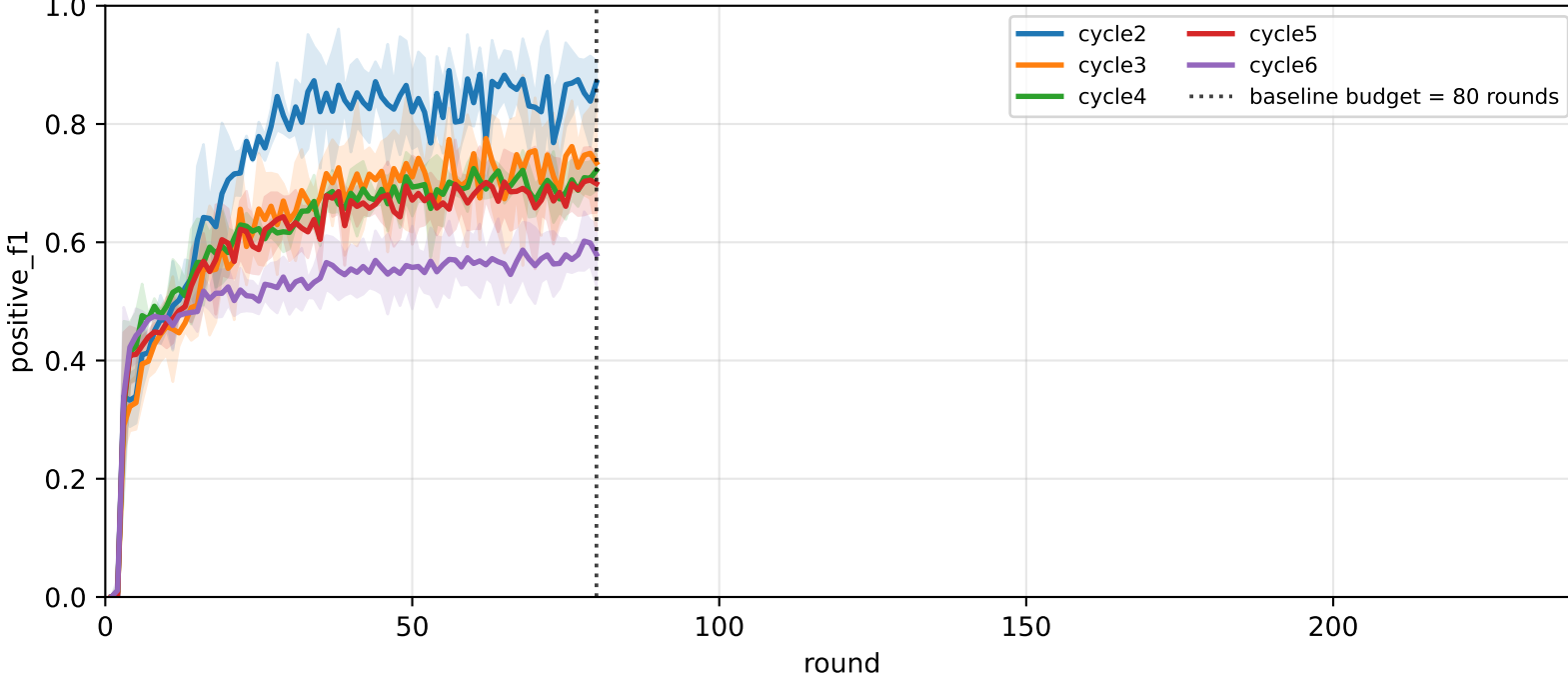
Validation loss comparison (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)



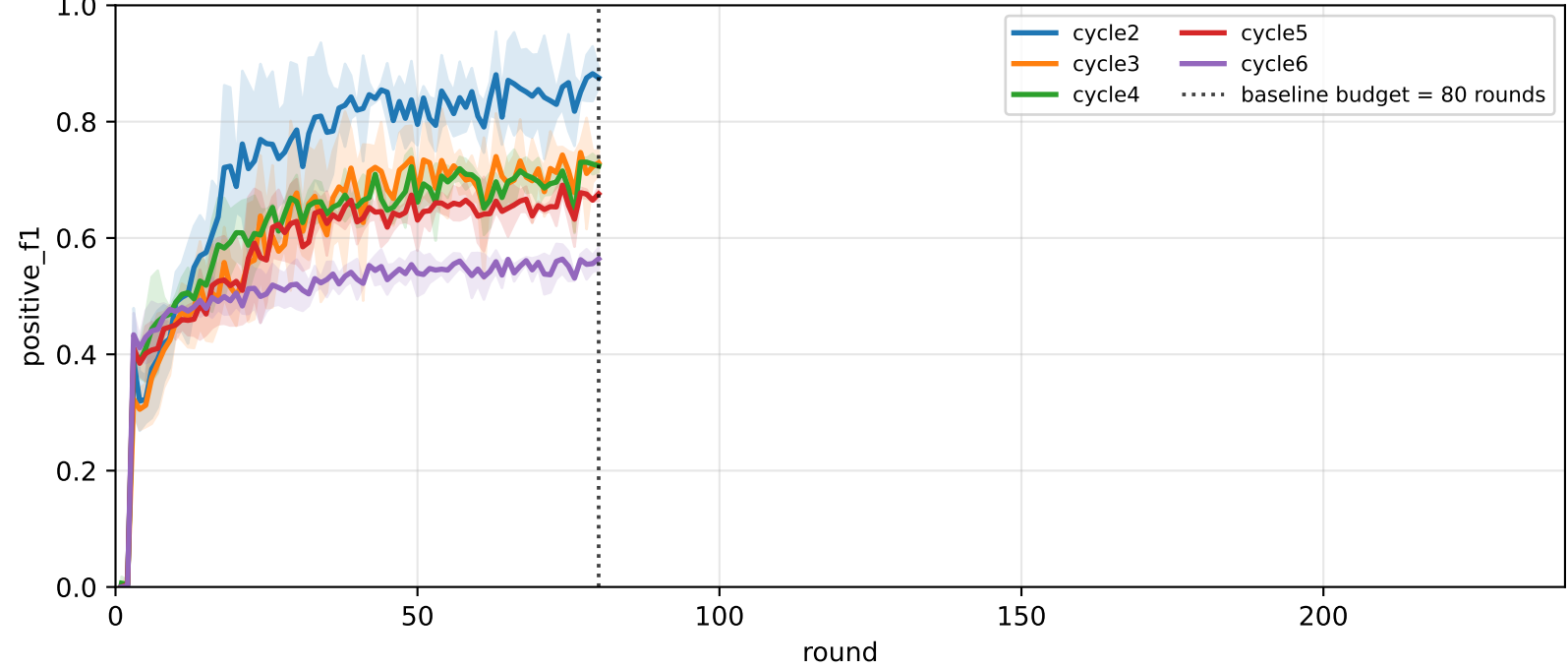
Pooled local centralized positive F1



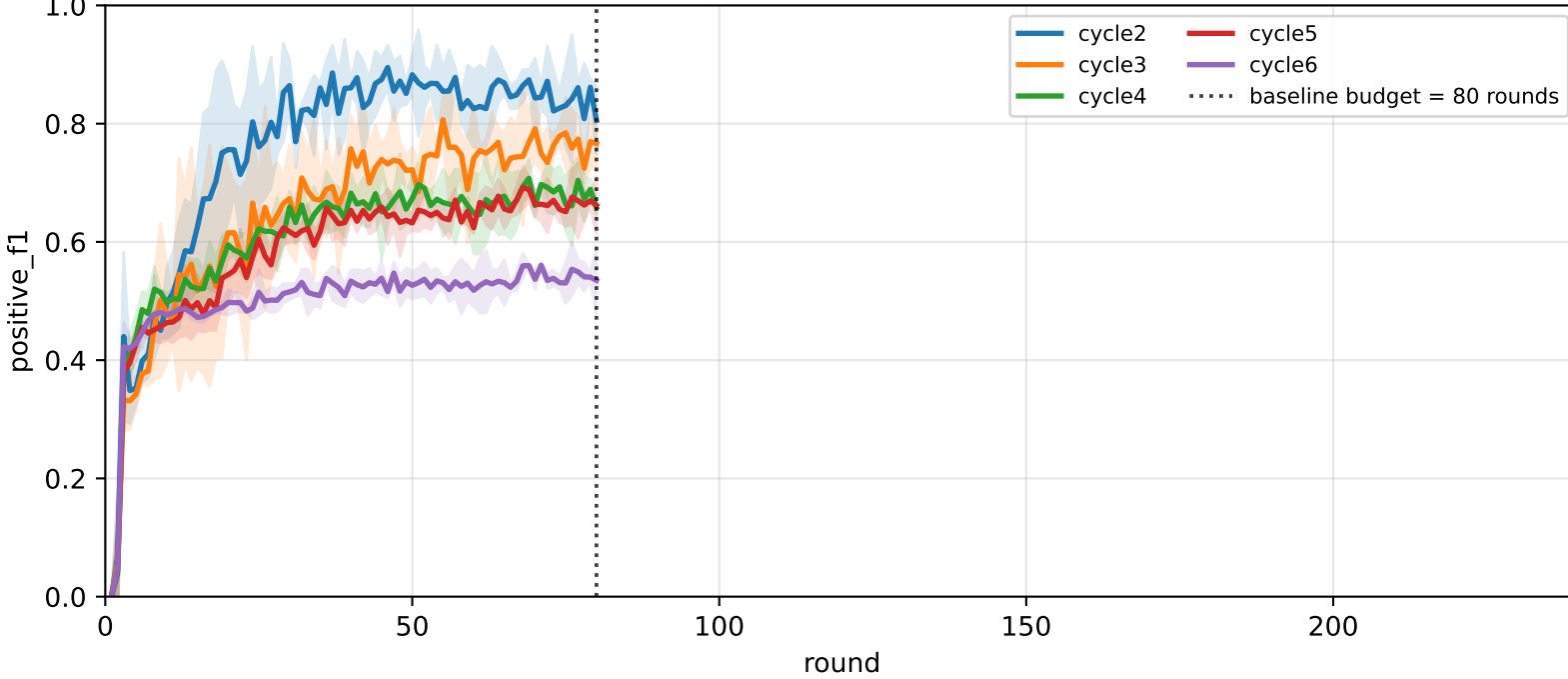
Pooled FedAvg positive F1



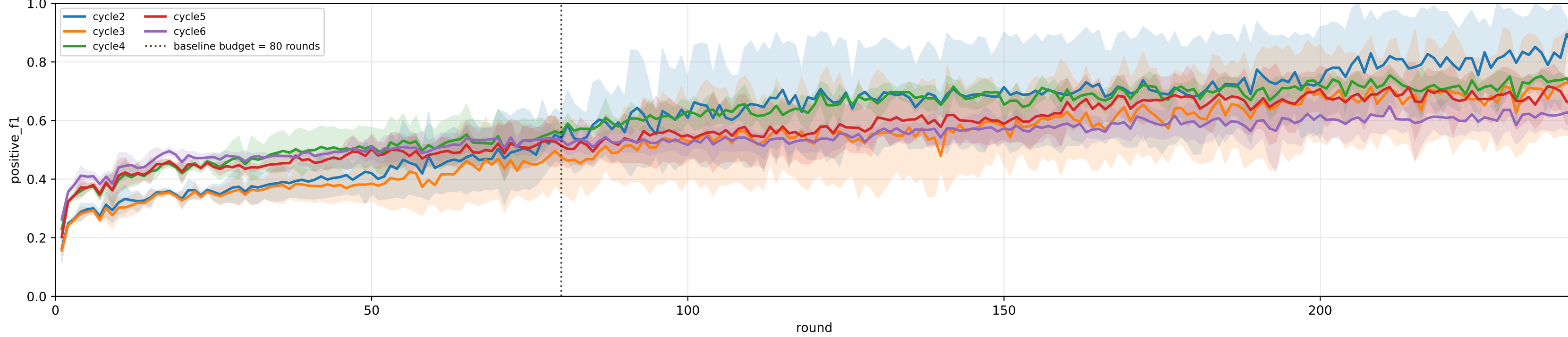
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, equal 240-round budget)



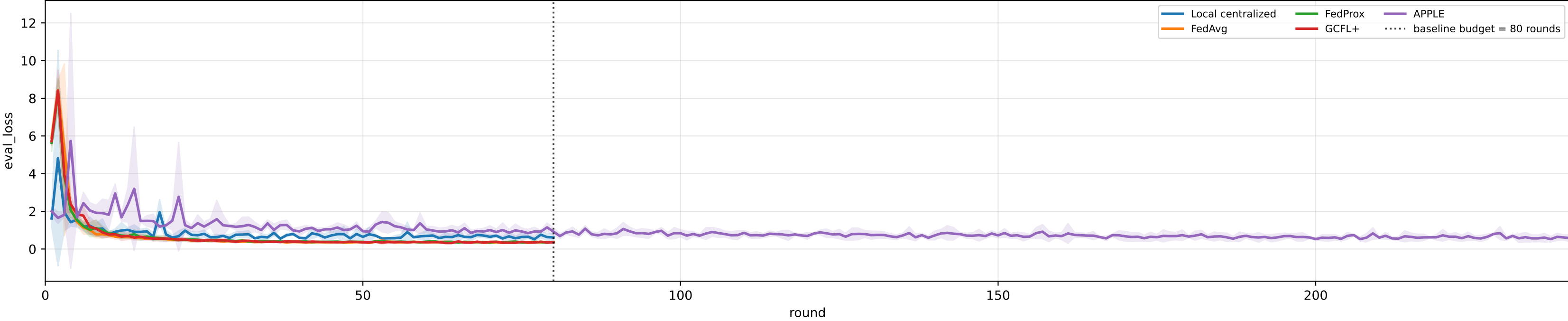
Controlled q-label heterogeneity: iid | family aggregated / pooled performance |
family=q_task_cycle6 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

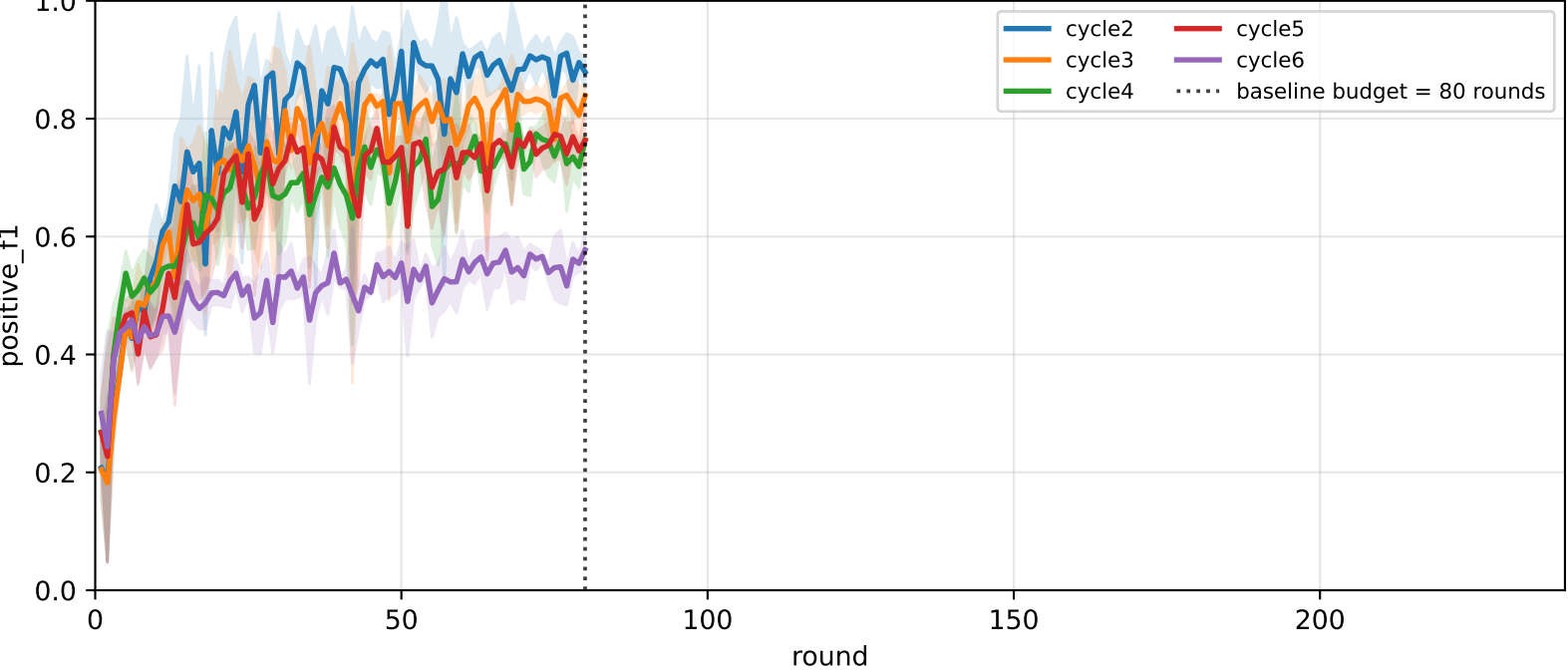
Best logged test performance (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	68.1% ± 1.5%	72.6% ± 1.2%	89.4% ± 1.7%	86.4% ± 0.7%	69.7% ± 3.6%	67.1% ± 2.2%	50.4% ± 1.9%
FedAvg	70.0% ± 2.6%	72.5% ± 2.6%	85.9% ± 3.4%	80.0% ± 5.1%	71.3% ± 1.6%	67.8% ± 4.1%	57.7% ± 3.3%
FedProx	65.9% ± 1.5%	68.9% ± 1.2%	83.9% ± 1.5%	76.8% ± 1.4%	67.9% ± 4.5%	63.4% ± 0.7%	52.6% ± 1.9%
GCFL+	66.9% ± 0.8%	70.1% ± 0.9%	82.6% ± 0.6%	83.0% ± 0.9%	68.4% ± 3.0%	63.8% ± 2.2%	52.6% ± 0.4%
APPLE	68.0% ± 4.9%	70.0% ± 4.8%	80.2% ± 11.6%	76.3% ± 11.8%	69.0% ± 2.1%	65.5% ± 6.1%	58.8% ± 3.5%

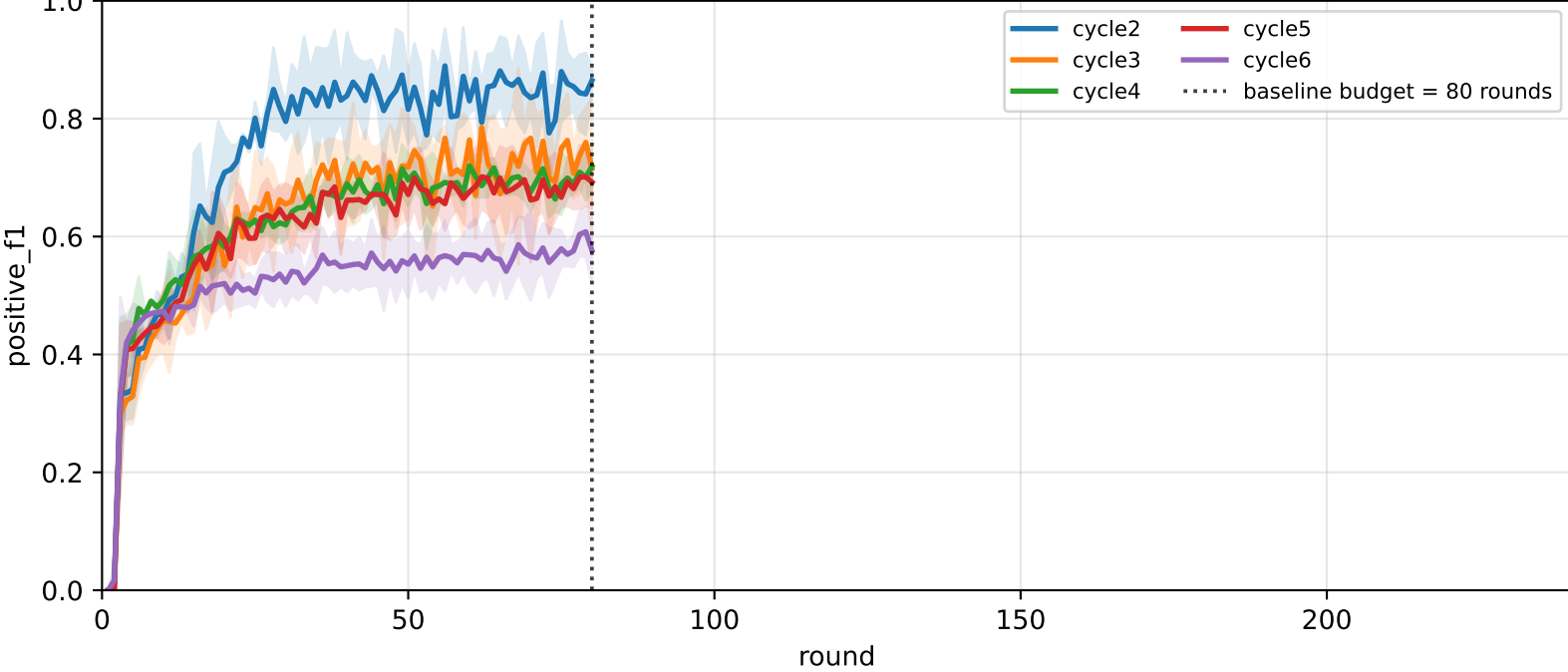
Validation loss comparison (all methods 80 rounds; APPLE dr_lr=0.001, mu=0.01, Lfrac=0.3)



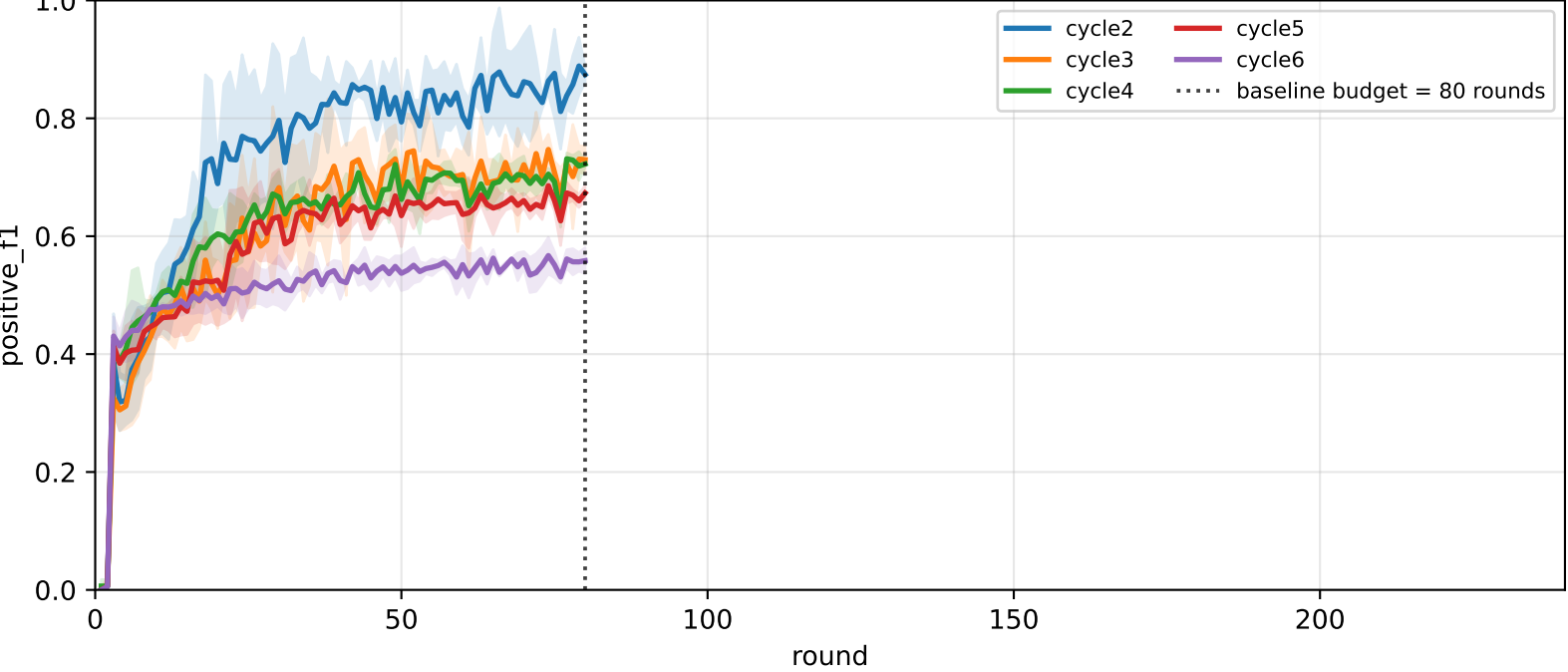
Pooled local centralized positive F1



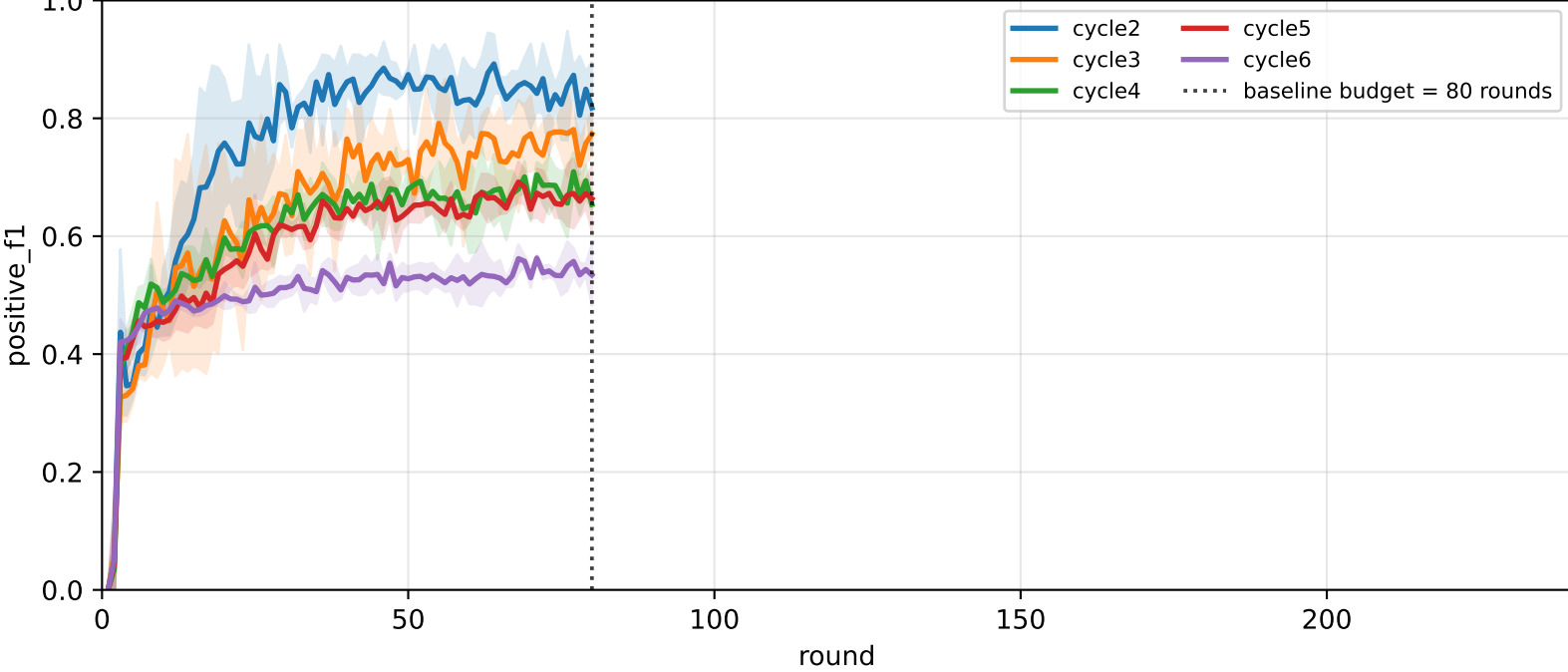
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, equal 240-round budget)

